

# Matthew Hurtado

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## Summary

Aspiring Backend Engineer with a strong foundation in Python, Java, and C++. Experienced in guiding students through complex mathematical and algorithmic challenges, demonstrating problem-solving skills applicable to backend development. Eager to leverage technical expertise in programming and machine learning to contribute to innovative backend projects.

## Education

### Rensselaer Polytechnic Institute (RPI)

Dec 2025

*Bachelor of Science, Computer Science & Mathematics*

- **GPA:** 3.76
- **Achievements:** Rensselaer Leadership Award, Chess Club, BJJ Club
- **Coursework:** Quantum Computing, Computational Optimization, Analysis, Linear Algebra

## Involvement and Experience

### RPI Computer Science Department

Aug 2023 - Present

*Teaching Assistant*

Troy, NY

- Assisted students in Computer Science I to develop problem-solving skills and programming etiquette, resulting in improved student performance and engagement
- Assisted students in Foundations of Computer Science to enhance proof-writing maturity and intuition, leading to a deeper understanding of theoretical concepts
- Guided students in Mathematical Foundations of Machine Learning to strengthen their mathematical and algorithmic understanding of ML concepts, which improved their ability to apply these concepts in practical scenarios
- Conducted recitation sessions to reinforce course material, resulting in increased student comprehension and retention of key topics

### RPI Advising and Learning Assistance Center

Aug 2024 - Present

*Tutor*

Troy, NY

- Assisted several students weekly in multivariable calculus and differential equations, leading to improved understanding and grades
- Assisted students in upper-level courses, including algorithms, numerical computing, and linear algebra, resulting in enhanced problem-solving skills
- Helped students build a strong foundation in problem-solving techniques and advanced mathematical concepts, contributing to their academic success

## Relevant Projects

### Parallel Discrete-Event Simulation of Pandemic Dynamics using ROSS and MPI

Jan 2025 - May 2025

- Co-authored a research paper detailing a high-performance parallel discrete-event simulation (PDES) of an agent-based model for pandemic dynamics.
- Designed and documented the model architecture and formulated the performance evaluation methodology for the simulation utilizing Rensselaer's Optimistic Simulation System (ROSS) and MPI.

### Financial Market Prediction with Autoregressive Models

Jun 2024 - Nov 2024

- Implemented transformer-based models to predict financial market closing prices.
- Compared autoregressive models, including linear regression, ARMA, RNNs, and transformers, for time-series prediction.
- Developed educational modules and published code on GitHub to teach autoregressive modeling techniques.

### Soft-margin Support Vector Machine

Jan 2024 - May 2024

- Reformulated the problem with the Augmented Lagrangian function, derived expressions for the iterative update scheme.
- Implemented the ALM in MATLAB, achieved >90% accuracy results on various data sets.

## Skills

- Python, Java, C++, C, LaTeX, MATLAB, MPI, CUDA, PDES, MIPS, Q#, Qiskit, SQL, C#, PyTorch, Pandas, NumPy, scikit-learn, TA-Lib